

Notes: Albanesi and Armenter (RES, 2012)

Intertemporal Distortions in the Second Best

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1 Big picture

This note summarizes Albanesi and Armenter (2012), which studies when second-best economies should feature permanent intertemporal distortions. The paper builds a unified framework that includes Ramsey taxation, limited commitment, self-enforcement, private information, political economy, and dynamic contracting environments. Its central result is a general sufficient condition: if the set of admissible allocations contains an allocation that eventually converges to the limiting first-best allocation, then intertemporal distortions are temporary in the second best.

One-sentence summary. The paper studies the long-run behaviour of the intertemporal wedge in a broad class of second-best dynamic economies, shows that permanent intertemporal distortions are ruled out whenever all distortions can be front-loaded, and uses this principle to unify results from Ramsey taxation, limited commitment, self-enforcement, private information, and firm-dynamics models.

Core result. The key condition is not a property of the optimal allocation itself. It is a property of the feasible and admissible set: if there exists at least one admissible allocation that converges to the limiting first-best allocation, then the second-best allocation cannot have a permanent one-sided intertemporal wedge. In deterministic economies, this typically means that the long-run Euler equation is undistorted.

Main intuition. A permanent intertemporal distortion is useful only if shifting resources across time relaxes future binding admissibility constraints. If all relevant future constraints can eventually be made slack by front-loading distortions, then future constraints no longer justify a wedge in the Euler equation. At that point, any permanent wedge would generate a first-order welfare gain from reallocating consumption across periods, so it cannot be optimal.

Important note on figures and tables. This is a theory paper and contains no empirical tables or original figures. Accordingly, the notes focus on the paper's mathematical objects, theorem, proof logic, and applications rather than extracted empirical tables. The titles of the equation/proposition blocks are placed in subsection titles so they appear in the Contents page.

2 Incremental contribution of the paper

First, the paper unifies many second-best environments. Instead of proving separate long-run results for each model, the paper develops a framework in which Ramsey taxation, limited commitment, self-enforcement constraints, private information, political economy, incomplete markets, and dynamic principal-agent problems can be represented as allocations subject to general admissibility constraints.

Second, the paper identifies the common principle behind temporary intertemporal distortions. In many Ramsey models, the Chamley-Judd logic implies that capital-income taxation or the intertemporal wedge vanishes in the long run. In other environments, such as private-information economies, permanent intertemporal distortions can arise. The paper explains the difference through the ability to front-load distortions: when all distortions can be shifted to the transition path, long-run intertemporal distortions are unnecessary.

Third, the paper separates intertemporal and intratemporal distortions. The sufficient

condition rules out permanent intertemporal wedges, but it does not imply that the second-best allocation converges fully to the first-best allocation. Labour wedges, risk-sharing distortions, and other intratemporal distortions may remain because they reflect the history of past binding constraints.

Fourth, the paper clarifies the role of constraints rather than instruments. A central message is that what matters is not whether an economy is implemented with capital taxes, asset taxes, trading restrictions, or other instruments. The robust object is the allocation and the admissibility constraints that characterize feasible second-best allocations. This makes the analysis more general than standard policy-instrument results.

Fifth, the paper provides a diagnostic tool. To know whether an environment can sustain permanent intertemporal distortions, one can ask whether there exists an admissible allocation that eventually behaves like the limiting first best. This may be easier than solving the full second-best problem.

3 Research question and main answer

The paper asks: **When are intertemporal distortions permanent in second-best economies?**

A first-best allocation equates the marginal utility value of resources across time. In a second-best economy, additional constraints may create an intertemporal wedge. This wedge can be interpreted as a distortion in savings, capital accumulation, or intertemporal consumption smoothing.

The paper's answer is: **intertemporal distortions are temporary whenever all future constraints can eventually be made nonbinding by front-loading distortions.** If there is an admissible path that converges to the limiting first-best allocation, the second-best allocation cannot display a permanent strictly positive or strictly negative intertemporal wedge.

The result does not say that all second-best economies converge to the first best. Rather, it says that the intertemporal Euler equation is not permanently distorted. Other margins can remain distorted forever.

4 Core concepts

4.1 First best and second best

The first best is the allocation that maximizes social welfare subject only to feasibility. Feasibility includes resource constraints, capital accumulation, and the consistency between individual and aggregate capital.

The second best adds extra constraints. These may be tax-implementability constraints, incentive compatibility constraints, self-enforcement constraints, limited-commitment constraints, or restrictions on asset markets and government instruments. The paper calls these extra restrictions **admissibility constraints**.

4.2 Intertemporal distortion

For an allocation ψ , the paper defines the intertemporal wedge for agent type i at node s^t as

$$\Delta_i(s^t) = u_i^c(s^t) - \beta \sum_{s^{t+1}|s^t} \pi(s^{t+1}|s^t) \nu_i^c(s^{t+1}) F_k(s^{t+1}), \quad (1)$$

where ν_i^c is marginal utility of consumption and F_k is the marginal product of capital.

At the first best,

$$\Delta_i(s^t) = 0 \quad (2)$$

for every type and every history. A nonzero $\Delta_i(s^t)$ means that the marginal value of consumption today differs from the discounted expected marginal value of saving and consuming tomorrow.

4.3 Front-loading distortions

Front-loading means placing distortions early in the allocation so that future admissibility constraints stop binding. Once constraints are slack, intertemporal reallocations are feasible. If there were a permanent intertemporal wedge after that point, shifting consumption across time would improve welfare.

This front-loading logic appears in several literatures, including Ramsey taxation and self-enforcing agreements. The paper shows that the same principle is the common reason why many different models have temporary intertemporal distortions.

4.4 Admissibility constraints

Admissibility constraints are the paper's general way of writing all restrictions beyond feasibility. They may include:

- implementability constraints in Ramsey taxation;
- self-enforcement constraints in risk-sharing models;
- incentive compatibility constraints in private-information economies;
- limited-commitment constraints on a government;
- constraints on the law of motion of auxiliary variables such as government debt.

The framework is abstract because the goal is not to solve one model but to identify a general structure shared by many second-best problems.

5 Examples motivating the framework

5.1 Ramsey taxation

In the Ramsey model, the government must finance expenditures using distortionary taxes because lump-sum taxes are ruled out. The core allocation restriction is an implementability constraint. In

a deterministic representative-agent model, this takes the form

$$\sum_{t=0}^{\infty} \beta^t (u_t^c c_t + u_t^l l_t) \geq u_0^c \{[(1 - \tau_0^k) r_0 + 1 - \delta] k_0 + b_0\}. \quad (3)$$

If the government can save enough, it can eventually finance expenditures from asset income, eliminate distortionary taxes, and converge to the first-best steady state. Therefore, the sufficient condition holds and intertemporal distortions are temporary.

5.2 Ramsey taxation with incomplete markets

With aggregate uncertainty and only risk-free government debt, the government faces a sequence of implementability constraints:

$$E_t \left\{ (u_t^c)^{-1} \sum_{d=t}^{\infty} \beta^{d-t} (u_d^c c_d + u_d^l l_d) - R_t k_t \right\} = b_{t-1} + V_t. \quad (4)$$

Here the auxiliary variables include government debt and transfers. If the government can accumulate enough assets, the sufficient condition can still hold. If there is an ad hoc asset limit, the condition can fail and long-run intertemporal distortions may arise.

5.3 Risk sharing with self-enforcement

In risk-sharing economies, agents may choose to leave the arrangement and take an outside option. A typical self-enforcement constraint is

$$E_t \left\{ \sum_{j=0}^{\infty} \beta^j u(c(\theta^{t+j}), l(\theta^{t+j})/\theta_{t+j}) \right\} \geq V^{out}(k_t; \theta_t). \quad (5)$$

If the first-best continuation allocation gives each agent at least the value of the outside option, then the sufficient condition holds. If the outside option becomes too attractive, the first-best limiting allocation is not admissible and permanent distortions may be possible.

5.4 Risk sharing with private information

With private information, allocations must satisfy incentive compatibility. A simplified form is

$$E_t \left\{ \sum_{j=0}^{\infty} \beta^j u(c(\theta^{t+j}), l(\theta^{t+j})/\theta_{t+j}) \right\} \geq E_t \left\{ \sum_{j=0}^{\infty} \beta^j u(c(\tilde{\theta}^{t+j}), l(\tilde{\theta}^{t+j})/\theta_{t+j}) \right\}. \quad (6)$$

The first-best allocation usually does not satisfy incentive compatibility because full insurance would induce agents to misreport their type. Hence, the sufficient condition typically fails. This is why private-information economies often display permanent intertemporal distortions.

5.5 Limited commitment by a benevolent government

If a benevolent government is subject to limited commitment, the allocation must give the government at least the value of its outside option:

$$U(x|s^t; s^t) \geq V^{out}(x(s^t), k(z^{t-1}), s^t). \quad (7)$$

Because a benevolent government values social welfare, the first-best continuation allocation is always at least as good as any feasible outside option. Therefore, limited commitment by a benevolent government alone does not generate permanent intertemporal distortions.

6 General framework

6.1 Environment

The economy is infinite horizon and stochastic. Agents may differ ex ante and ex post. Aggregate histories are denoted by z^t , idiosyncratic histories by θ^t , and full histories by $s^t = (z^t, \theta^t)$.

The key allocation objects are:

- consumption $c_i(s^t)$,
- labour $l_i(s^t)$,
- individual capital holdings $y_i(s^{t-1})$,
- aggregate capital $k(z^t)$.

The flow allocation is

$$x_i(s^t) = \{c_i(s^t), l_i(s^t)\}. \quad (8)$$

An allocation is

$$\psi = \{x, y, k\}. \quad (9)$$

6.2 Feasibility

The aggregate resource constraint is

$$c(z^t) + k(z^t) + g(z^t) \leq F(k(z^{t-1}), l(z^t); z^t), \quad (10)$$

where aggregate consumption and labour are weighted averages across agents and idiosyncratic types.

The consistency condition for individual and aggregate capital is

$$\sum_{i \in I} \pi_i \left(\sum_{\theta^t \in \Theta^t} \pi_{\theta}(\theta^t) y_i(z^t, \theta^t) \right) \leq k(z^t). \quad (11)$$

A first-best allocation maximizes social welfare subject only to these feasibility conditions.

6.3 General admissibility constraint

The paper writes the extra constraints beyond feasibility as

$$H_{m,i}(\psi, s^t) \leq a_{m,i}(s^t), \quad (12)$$

where m indexes the type of constraint, i indexes the ex ante type, and $a_{m,i}(s^t)$ is an auxiliary variable.

The function $H_{m,i}$ has the form

$$H_{m,i}(\psi, s^t) = b_{m,i}(\psi, s^t) + h_{m,i}^0(\psi, s^t) + \sum_{\tilde{s}^t \in S^t | s^{t-1}} h_{m,i}^1(\psi, s^t; \tilde{s}^t). \quad (13)$$

This specification is flexible enough to include constraints on current states, continuation utility, and cross-type deviations.

6.4 Auxiliary variable and its law of motion

The auxiliary variable $a(s^t)$ may represent endogenous state variables such as government debt, continuation promises, or other objects needed to parameterize future constraints. Its law of motion is restricted by

$$a(s^t) \in \Gamma(a(s^{t-1}); s^t). \quad (14)$$

This allows the model to handle restrictions on asset positions, borrowing, government debt, and other state variables.

7 Main theorem

7.1 Condition 1: existence of an admissible path to the limiting first best

The sufficient condition is:

There exists an admissible allocation $\tilde{\psi}$ such that the associated sequence of probability measures $\{\tilde{\mu}_t\}$ converges weakly to μ_∞^{fb} , the limiting first-best probability measure, for any initial conditions.

In plain English, there must exist a feasible and admissible path that eventually looks like the first-best limiting allocation. It does not need to be the optimal path. It only needs to exist.

7.2 Theorem 1: no permanent one-sided intertemporal wedge

Let ψ^* be a second-best allocation and μ^* its associated sequence of probability measures. If Condition 1 holds, then for each ex ante type i , either

$$\mu_\infty^*(\Delta_i(s^t) = 0) = 1, \quad (15)$$

or the intertemporal wedge fluctuates around zero:

$$\mu_{\infty}^*(\Delta_i(s^t) > 0) > 0, \quad \mu_{\infty}^*(\Delta_i(s^t) < 0) > 0. \quad (16)$$

Interpretation. The theorem rules out a permanent strictly positive or strictly negative intertemporal distortion. In deterministic steady-state economies, the result implies a zero long-run intertemporal wedge. In stochastic economies, the wedge may fluctuate, but it cannot remain one-sided forever in the limiting distribution.

7.3 What the theorem does not say

The theorem does not say that:

- the second-best allocation converges fully to the first best;
- all constraints stop binding immediately;
- all intratemporal wedges vanish;
- Condition 1 is necessary.

It only establishes a sufficient condition for temporary intertemporal distortions.

8 Proof logic

8.1 Stage 1: if future constraints stop binding, intertemporal wedges cannot persist

The first stage takes the plan for the auxiliary variable as given. The key lemma states that if the probability that all future admissibility constraints are slack converges to one, then the long-run intertemporal wedge is either zero or changes sign.

The intuition is straightforward. Suppose all future constraints are slack. Then shifting a small amount of consumption from today to tomorrow does not violate admissibility constraints. If the marginal utility value today differs from the discounted expected marginal utility value tomorrow, such a shift generates a first-order welfare gain. Therefore, a permanent one-sided wedge cannot be optimal.

8.2 Stage 2: Condition 1 implies future admissibility constraints eventually stop binding

The second stage studies the optimal path of the auxiliary variable. Condition 1 implies that the set of auxiliary variables that can support the first-best limiting allocation is nonempty. The paper then shows that the optimal path can be chosen so that future admissibility constraints are eventually relaxed.

The proof uses the convexity and continuity of the correspondence Γ governing the auxiliary variable. It also uses the regularity condition that rules out local maxima and inflection points.

8.3 Combining the stages

Condition 1 gives the ability to front-load all distortions. Once distortions are front-loaded, future constraints stop binding. Once future constraints stop binding, the intertemporal margin must be undistorted in the limit. This completes the logic behind Theorem 1.

9 Application 1: Ramsey taxation

9.1 Why the sufficient condition usually holds

In the standard Ramsey model, the government can use distortionary taxes and debt. If it can save enough, it can accumulate sufficient assets so that interest income eventually finances government expenditures. After that point, distortionary taxes can be eliminated and the economy can converge to the first-best steady state.

Therefore, there exists an admissible allocation that converges to the limiting first best. Condition 1 holds, and intertemporal distortions are temporary.

9.2 Connection to the Chamley-Judd result

The Chamley-Judd result says that capital income taxes are zero in the long run. Albanesi and Armenter's theorem explains this result as part of a broader principle: the government can front-load distortions by taxing early, accumulating assets, and eventually removing the need for intertemporal distortions.

9.3 When Ramsey models can have long-run intertemporal distortions

If the government faces an ad hoc asset limit, or if private agents face borrowing constraints that prevent the government from accumulating enough assets, then the sufficient condition may fail. In that case, permanent intertemporal distortions may arise.

A balanced-budget Ramsey model is more subtle. The sufficient condition fails because the government cannot save. Yet if government spending is small enough, there may still be no long-run intertemporal wedge because it is enough to front-load some distortions, not necessarily all distortions.

10 Application 2: risk sharing

10.1 Self-enforcement constraints

In self-enforcement economies, agents must prefer staying in the risk-sharing arrangement to taking an outside option. Whether Condition 1 holds depends on the outside option and parameters.

If the outside option is not too attractive, the first-best limiting allocation can satisfy the self-enforcement constraints. Then distortions can be front-loaded, and intertemporal distortions are

temporary.

If the outside option is too attractive, the first-best limiting allocation is not admissible. Then permanent distortions may be possible.

10.2 Private information

With private information, the first-best allocation typically violates incentive compatibility because agents would misreport their type under full insurance. The sufficient condition usually fails.

In standard private-information economies, future incentive constraints keep binding. The optimal allocation front-loads consumption for incentive reasons, generating a permanent intertemporal wedge. This explains why Golosov, Kocherlakota, and Tsyvinski type environments often imply permanent intertemporal distortions.

10.3 Why private information differs from self-enforcement

Both self-enforcement and private information involve constraints on continuation allocations, but the sign of the front-loading logic differs.

In self-enforcement models, promising more future utility can relax participation constraints, so consumption may be back-loaded. In private-information models, promising future consumption tightens future incentive constraints, so consumption is front-loaded. This is why private-information economies often retain permanent intertemporal distortions.

11 Application 3: government limited commitment

11.1 Benevolent government

If the government is benevolent, limited commitment alone cannot generate permanent intertemporal distortions. The reason is that the government's outside option is feasible, while the first-best continuation maximizes social welfare over all feasible continuations. Therefore, the first-best continuation is at least as good as the outside option.

Condition 1 holds, so intertemporal distortions are temporary.

11.2 Self-interested government

If the policymaker is self-interested, as in political economy models, the result depends on the severity of the conflict between private agents and the ruler. If the ruler's outside option is weak, the discount factor is high, or the ruler receives a sufficiently high Pareto weight, then the first-best limiting allocation can satisfy the government's participation constraint. Condition 1 holds.

If the ruler can extract large rents from output, then the first-best limiting allocation may fail to satisfy the constraint. In that case, permanent intertemporal distortions may arise.

12 Application 4: firm dynamics

12.1 Optimal lending contracts

In models such as Albuquerque and Hopenhayn (2004), borrowing constraints arise from self-enforcement. The optimal contract front-loads distortions: as the borrower accumulates equity, borrowing constraints relax and eventually stop binding. Investment converges toward the first-best level.

12.2 Moral hazard and firm dynamics

In models such as Clementi and Hopenhayn (2006), incentive constraints may relax after sufficiently good histories. If the entrepreneur becomes sufficiently well-capitalized, incentive constraints stop binding and capital reaches the first-best level. The structure again reflects the front-loading principle.

13 Core mathematical objects and results

13.1 Table A: Main mathematical objects; Table B: Main logical implications

Object	Meaning	Condition or model	Implication
$\psi = \{x, y, k\}$	Allocation: flow allocation, individual capital, aggregate capital	Admissible path to limiting first best exists	No permanent one-sided intertemporal wedge
ψ^{fb}	First-best allocation subject only to feasibility	Government can save in Ramsey model	Distortions can be front-loaded
ψ^*	Second-best allocation subject to feasibility and admissibility	Ad hoc asset limits bind	Permanent wedge may arise
$H_{m,i} \leq a_{m,i}$	General admissibility constraint	Self-enforcement outside option weak	Condition 1 can hold
$a(s^t) \in \Gamma(a(s^{t-1}); s^t)$	Law of motion for auxiliary variables	Private information with risk aversion	Condition 1 usually fails
$\Delta_i(s^t)$	Intertemporal wedge for type i	Benevolent government with limited commitment	Limited commitment alone does not create permanent wedge
μ_∞^{fb}	Limiting first-best probability measure		

(a) Table A: Main mathematical objects

(b) Table B: Main logical implications

Summary tables constructed from the paper's theoretical framework

13.2 Equation block 1: Ramsey implementability and front-loading

In the deterministic Ramsey model, admissibility is summarized by the implementability constraint

$$\sum_{t=0}^{\infty} \beta^t (u_t^c c_t + u_t^l l_t) \geq u_0^c \{[(1 - \tau_0^k) r_0 + 1 - \delta] k_0 + b_0\}. \quad (17)$$

The government can front-load distortions by raising revenues early and accumulating assets. Once government assets are high enough, taxes can be eliminated and the limiting first-best allocation becomes admissible.

13.3 Equation block 2: Admissibility constraint and auxiliary variable

The generic admissibility restriction is

$$H_{m,i}(\psi, s^t) \leq a_{m,i}(s^t), \quad (18)$$

with law of motion

$$a(s^t) \in \Gamma(a(s^{t-1}); s^t). \quad (19)$$

This is the mathematical device that lets the paper cover many different second-best environments in one notation.

13.4 Equation block 3: Intertemporal wedge

The intertemporal wedge is

$$\Delta_i(s^t) = u_i^c(s^t) - \beta \sum_{s^{t+1}|s^t} \pi(s^{t+1}|s^t) u_i^c(s^{t+1}) F_k(s^{t+1}). \quad (20)$$

The first-best Euler equation corresponds to $\Delta_i(s^t) = 0$. Theorem 1 says that if Condition 1 holds, the limiting wedge cannot be permanently positive or permanently negative.

13.5 Equation block 4: Condition 1 and Theorem 1

Condition 1 requires the existence of an admissible allocation $\tilde{\psi}$ such that

$$\tilde{\mu}_t \Rightarrow \mu_{\infty}^{fb}. \quad (21)$$

If this holds, then for each type i , either

$$\mu_{\infty}^*(\Delta_i = 0) = 1, \quad (22)$$

or

$$\mu_{\infty}^*(\Delta_i > 0) > 0, \quad \mu_{\infty}^*(\Delta_i < 0) > 0. \quad (23)$$

This is the paper's main theorem.

14 How to read the paper's sufficient condition

14.1 Why the condition is sufficient, not necessary

The condition requires that *all* distortions can be front-loaded. Some models need only partial front-loading to eliminate intertemporal distortions. For example, in the balanced-budget Ramsey model, the government cannot save and therefore cannot front-load all distortions. But if government spending is small enough, capital can still converge to the first-best steady-state level, so the intertemporal wedge can vanish.

Therefore, failure of Condition 1 does not automatically mean permanent intertemporal distortions. It means that the paper's general theorem does not apply, and a model-specific analysis is needed.

14.2 Why the condition is useful

The condition is useful because it is often easier to verify existence of an admissible path to the limiting first best than to solve the full second-best problem. It gives a way to infer long-run properties without full characterization.

14.3 Why the condition is not about the optimum

The existence of an admissible first-best-convergent path does not mean the optimum chooses that path. The second-best optimum may still differ from the first best in the limit on intratemporal margins. The theorem only says that permanent intertemporal wedges are not needed.

15 Relation to important literatures

15.1 Relation to Chamley-Judd capital taxation

Chamley-Judd results are special cases of the paper's broader theorem. In Ramsey models, capital taxes distort the intertemporal margin. If the government can front-load taxes and eventually self-finance, long-run capital taxes are unnecessary.

15.2 Relation to dynamic Mirrlees taxation

Dynamic Mirrlees environments typically feature private information. Incentive compatibility constraints do not vanish at the first-best limiting allocation. Thus, the sufficient condition usually fails, and permanent intertemporal distortions may arise.

15.3 Relation to self-enforcing agreements

Self-enforcement models often front-load distortions because future utility promises relax participation constraints. The paper links this front-loading logic to the absence of permanent intertemporal wedges.

15.4 Relation to political economy

Political economy constraints can be represented as admissibility constraints on government policy. If the policymaker is benevolent, limited commitment alone does not generate permanent intertemporal distortions. If the policymaker is self-interested, distributional conflict may prevent the sufficient condition from holding.

16 Interpretation and economic intuition

16.1 Why future binding constraints matter

A current binding constraint can distort the level of consumption or labour today. But it does not by itself justify a permanent wedge between today and tomorrow. A permanent intertemporal wedge requires that shifting resources across time changes the tightness of future constraints. Hence, the wedge is about future constraints, not simply past or current ones.

16.2 Why intratemporal distortions can persist

Even if future constraints stop binding, the Lagrange multipliers on past constraints remain embedded in the allocation. These past constraints can leave permanent effects on intratemporal margins. This is why the theorem does not imply complete convergence to the first best.

16.3 Why private information is different

Private information creates constraints that remain active because incentives must be provided in every future period. If agents must be induced to reveal their types, the planner cannot simply wait until tomorrow and implement the first best. This is why private-information economies are the canonical case in which intertemporal distortions can be permanent.

17 Limitations and assumptions

17.1 Generality comes at a cost

Because the framework is very general, the paper imposes assumptions directly on the second-best allocation: interiority, convergence, and regularity conditions ruling out problematic local maxima or inflection points. These assumptions are common in the literature but may need to be checked or replaced with primitive assumptions in specific models.

17.2 Condition 1 may be too strong for specific applications

The sufficient condition is designed to work across a broad class of environments. In a particular subclass, a weaker condition may be enough. The Ramsey balanced-budget example illustrates this clearly.

17.3 No overlapping generations

The framework uses infinitely lived agents. Overlapping-generations economies can feature dynamic inefficiency and intergenerational redistribution, which may create different rationales for intertemporal distortions.

18 Short summary

18.1 Core takeaway

The paper’s central message is:

Permanent intertemporal distortions are optimal only when they help relax future admissibility constraints that cannot be eliminated by front-loading distortions.

If an admissible allocation can eventually converge to the limiting first best, then future constraints can eventually be made slack. In that case, the second best cannot have a permanent one-sided intertemporal wedge.

18.2 Model-class summary

- **Ramsey taxation with enough government saving:** sufficient condition holds; no permanent intertemporal distortions.
- **Ramsey with ad hoc saving limits:** condition may fail; permanent wedges may arise.
- **Self-enforcement:** depends on outside option and parameters.
- **Private information:** condition usually fails; permanent wedges are common.
- **Benevolent limited-commitment government:** condition holds; limited commitment alone does not create permanent intertemporal distortions.
- **Self-interested political economy:** depends on severity of distributional conflict.
- **Firm dynamics with self-enforcement or moral hazard:** many optimal contracts front-load distortions and may eventually relax constraints.

19 Conclusion

Albanesi and Armenter (2012) provide a general theory of when intertemporal distortions disappear in the long run. The paper’s unified framework shows that many seemingly different results in macroeconomics share the same logic: if distortions can be front-loaded, then future constraints eventually stop binding, and a permanent intertemporal wedge cannot be optimal.

The result clarifies why Ramsey models often imply zero long-run capital taxation, why private-information models often imply permanent wedges, and why limited commitment by a benevolent

government alone is not enough to generate permanent intertemporal distortions. The paper's main contribution is therefore not a new solution to one model, but a general organizing principle for second-best dynamic economies.